

**Q:01**

```
t=-4:.01:20;  
y=2.5;  
plot(t,y)
```

**Q:02**

```
t=-4:.01:20;  
y=2.5;  
z=(10/pi)*sin(pi*t/4);  
x=y+z;  
plot(t,x)
```

**Q:03**

```
t=-4:.01:20;  
y=2.5;  
z=(10/pi)*sin(pi*t/4);  
p=(10/pi)*(1/3)*sin(3*pi*t/4);  
x=y+z+p;  
plot(t,x)
```

**Q-04**

```
t=-4:.01:20;  
y=2.5;  
z=(10/pi)*sin(pi*t/4);  
p=(10/pi)*(1/3)*sin(3*pi*t/4);  
q=(10/pi)*(1/5)*sin(5*pi*t/4);  
x=y+z+p+q;  
plot(t,x)
```

**Q-05**

```
t=-4:.01:16;  
y=2.5;  
z=(10/pi)*sin(pi*t/4);  
p=(10/pi)*(1/3)*sin(3*pi*t/4);  
q=(10/pi)*(1/5)*sin(5*pi*t/4);  
r=(10/pi)*(1/7)*sin(7*pi*t/4);  
x=y+z+p+q+r;  
plot(t,x)
```

**Q-06**

```
t=-4:.01:16;  
y=2.5;  
z=(10/pi)*sin(pi*t/4);  
p=(10/pi)*(1/3)*sin(3*pi*t/4);  
q=(10/pi)*(1/5)*sin(5*pi*t/4);  
r=(10/pi)*(1/7)*sin(7*pi*t/4);  
s=(10/pi)*(1/9)*sin(9*pi*t/4);  
x=y+z+p+q+r+s;
```

plot(t,x)

### Q-07

t=-4:.01:16;

y=2.5;

z=(10/pi)\*sin(pi\*t/4);

p=(10/pi)\*(1/3)\*sin(3\*pi\*t/4);

q=(10/pi)\*(1/5)\*sin(5\*pi\*t/4);

r=(10/pi)\*(1/7)\*sin(7\*pi\*t/4);

s=(10/pi)\*(1/9)\*sin(9\*pi\*t/4);

u=(10/pi)\*(1/11)\*sin(11\*pi\*t/4);

x=y+z+p+q+r+s+u;

plot(t,x)

### Q-08

t=-4:.01:16;

y=2.5;

z=(10/pi)\*sin(pi\*t/4);

p=(10/pi)\*(1/3)\*sin(3\*pi\*t/4);

q=(10/pi)\*(1/5)\*sin(5\*pi\*t/4);

r=(10/pi)\*(1/7)\*sin(7\*pi\*t/4);

s=(10/pi)\*(1/9)\*sin(9\*pi\*t/4);

u=(10/pi)\*(1/11)\*sin(11\*pi\*t/4);

v=(10/pi)\*(1/13)\*sin(13\*pi\*t/4);

x=y+z+p+q+r+s+u+v;

plot(t,x)

### Q-09 ( See Page # 67-72 from your lecture sheet)

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \underbrace{\left[ -\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4} \right]}_{\text{AC value}}$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \left[ -\frac{5}{\pi} \frac{1}{1} (\cos 1\pi - 1) \sin \frac{1\pi t}{4} - \frac{5}{\pi} \frac{1}{2} (\cos 2\pi - 1) \sin \frac{2\pi t}{4} - \frac{5}{\pi} \frac{1}{3} (\cos 3\pi - 1) \sin \frac{3\pi t}{4} - \frac{5}{\pi} \frac{1}{4} (\cos 4\pi - 1) \sin \frac{4\pi t}{4} + \dots \right]$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \left[ -\frac{5}{\pi} (-1 - 1) \sin \frac{\pi t}{4} - \frac{5}{\pi} \frac{1}{2} (1 - 1) \sin \frac{2\pi t}{4} - \frac{5}{\pi} \frac{1}{3} (-1 - 1) \sin \frac{3\pi t}{4} - \frac{5}{\pi} \frac{1}{4} (1 - 1) \sin \frac{4\pi t}{4} + \dots \right]$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \frac{10}{\pi} \sin \frac{\pi t}{4} - \frac{5}{\pi} \frac{1}{2} (0) \sin \frac{2\pi t}{4} + \frac{10}{3\pi} \sin \frac{3\pi t}{4} - \frac{5}{\pi} \frac{1}{4} (0) \sin \frac{4\pi t}{4} + \dots$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \frac{10}{\pi} \sin \frac{\pi t}{4} + \frac{10}{3\pi} \sin \frac{3\pi t}{4} + \frac{10}{5\pi} \sin \frac{5\pi t}{4} + \frac{10}{7\pi} \sin \frac{7\pi t}{4} + \dots$$

$$f(t) = 2.5 + \frac{10}{\pi} \left( \sin \frac{\pi t}{4} + \frac{1}{3} \sin \frac{3\pi t}{4} + \frac{1}{5} \sin \frac{5\pi t}{4} + \frac{1}{7} \sin \frac{7\pi t}{4} + \dots \right)$$

```

t=-4:.001:20;
y=2.5;
z=(10/pi)*sin(pi*t/4);
p=(10/pi)*(1/3)*sin(3*pi*t/4);
q=(10/pi)*(1/5)*sin(5*pi*t/4);
r=(10/pi)*(1/7)*sin(7*pi*t/4);
s=(10/pi)*(1/9)*sin(9*pi*t/4);
u=(10/pi)*(1/11)*sin(11*pi*t/4);
v=(10/pi)*(1/11)*sin(11*pi*t/4);
x=y+z+p+q+r+s+u+v;
result=[t;z;p;q;r;s;u;v;x];
plot(t,x)

```

**Q-10:** ( See Page # 55 from your lecture sheet)

$$f(x) = 1 + 2 \sin t - \sin 2t + \frac{2}{3} \sin 3t - \frac{1}{2} \sin 4t + \frac{2}{5} \sin 5t + \dots$$

```

t=-4:.01:16;
y=1;
w=1;
z1=2*sin(w*t);
p1=(-1)*sin(w*2*t);
p2=(2/3)*sin(w*3*t);
p3=(-1/2)*sin(w*4*t);
p4=(2/5)*sin(w*5*t);

```

```

z=z1+p1+p2+p3+p4;
q=z+y;
plot(t,q);

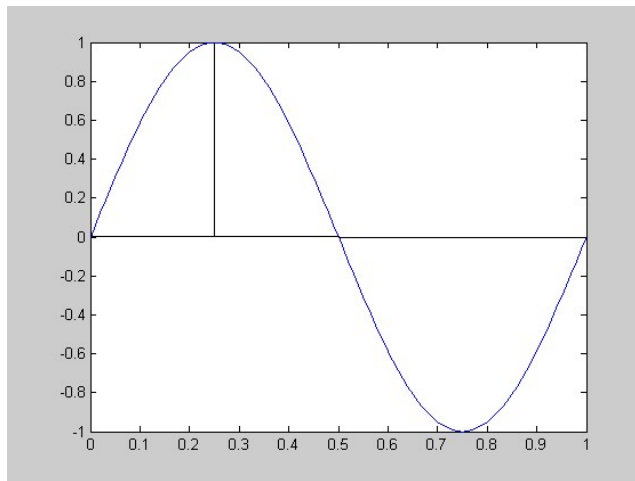
```

**Q-11:**

```

t=0:.01:1;
A=1
w=2*pi*1 %  $\omega = 2\pi f$ 
y=A*sin(w*t);
plot(t,y)

```



**Q-12:**

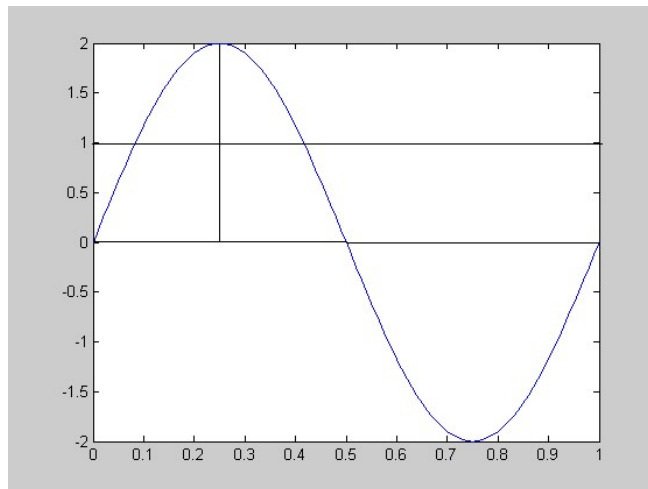
**t=0:.01:1;**

**A=2**

**w=2\*pi\*1 %  $\omega = 2\pi f$**

**y=A\*sin(w\*t);**

**plot(t,y)**



**Q-13:**

$$x = \frac{1}{4} \cos 8t + \frac{1}{8} \sin 8t$$

**A1=1/4;B1=1/8;**

**t=0:0.01:1;**

**w=8;**

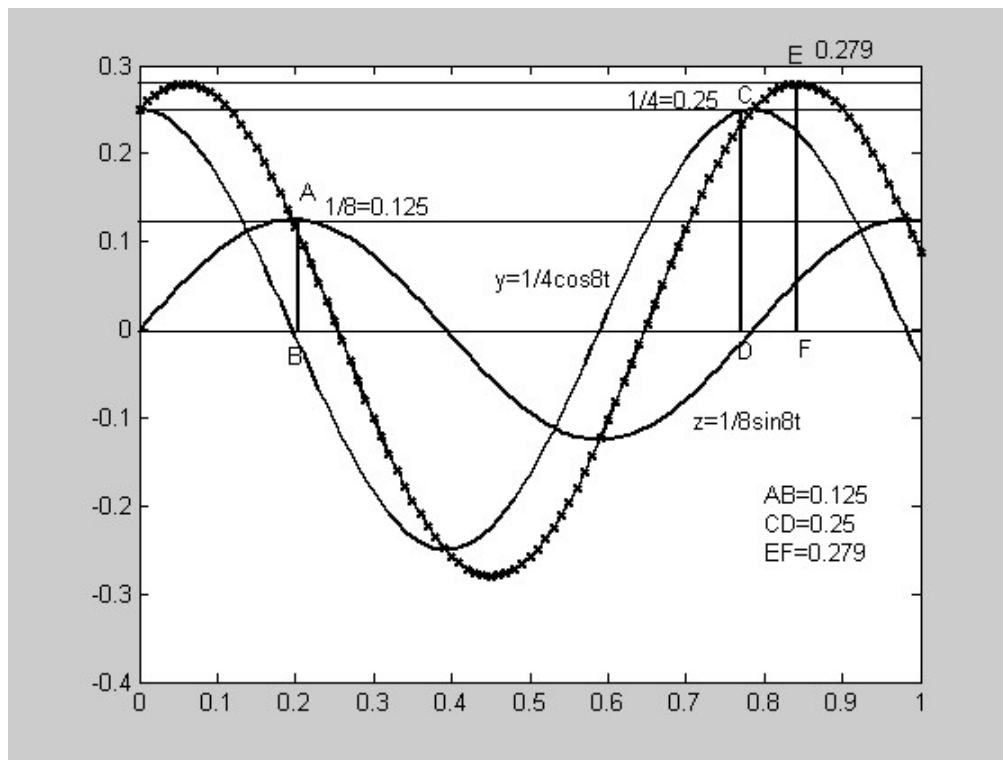
**y=A1\*cos(w\*t);**

**z=B1\*sin(w\*t);**

**x=y+z**

**result=[t;y;z;x];**

**plot(t,y,t,z,t,x)**



|   |      |           |          |          |          |          |                |
|---|------|-----------|----------|----------|----------|----------|----------------|
| t | 0    | 0.01      | 0.02     | 0.03     | 0.04     | 0.05     | 0.06           |
| y | 0.25 | 0.2492    | 0.24681  | 0.24283  | 0.23731  | 0.23027  | 0.22175        |
| z | 0    | 0.0099893 | 0.019915 | 0.029713 | 0.039321 | 0.048677 | 0.057722       |
| x | 0.25 | 0.25919   | 0.26672  | 0.27255  | 0.27663  | 0.27894  | <b>0.27947</b> |

We have,

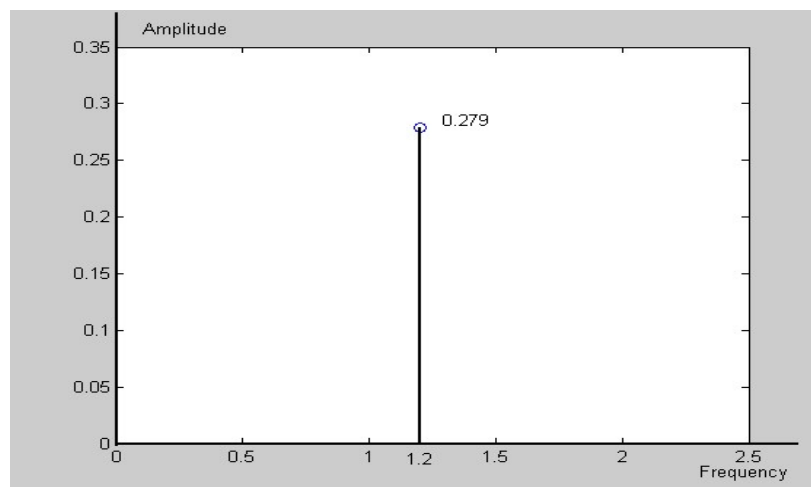
Hence The resulting line spectrum is:

$f=1:0.01:2;$

$f=[1.2]$

$a=[0.279]$

$\text{stem}(f,a)$



Q-14:

```
a=[ 2   6   3   1   5   6   9]
plot(a)
grid on
```

Q-15:

```
bar(a)
xlabel('Sample #')
ylabel('Pounds')
```

Q-16:

```
plot(b, '*')
axis([0 10 0 10])
```

Q-17:

```
A = [1 2 0; 2 5 -1; 4 10 -1]
%We can easily find the transpose of the matrix A.
B = A'
```

Q-18:

```
%Now let's multiply these two matrices together.
C = A * B
```

Q-19:

```
%Instead of doing a matrix multiply, we can multiply the corresponding elements of two
matrices or vectors using the .* operator.
```

```
C = A .* B
clc
```

Q-20:

```
%Let's find the inverse of a matrix ...
X = inv(A)
```

Q-21:

```
%a matrix times its inverse is the identity matrix.
I = inv(A) * A
```

Q-22:

```
%There are functions to obtain eigenvalues
eig(A)
```

Q-23:

```
%As well as the singular value decomposition.
svd(A)
```

Q-24:

%The "poly" function generates a vector containing the coefficients of the characteristic polynomial. The characteristic polynomial of a matrix A is  
`poly(A)`

Q-25:

`p = round(poly(A))`

Q-26:

We can easily find the roots of a polynomial using the roots function.  
These are actually the eigenvalues of the original matrix.  
`roots(p)`

Q-27:

MATLAB has many applications beyond just matrix computation.  
To convolve two vectors  
`q = conv(p,p)`

Q-28:

convolve again and plot the result  
`r = conv(p,q)`  
`plot(r);`

Q-29:

%At any time, we can get a listing of the variables we have stored in memory using the who or whos command.  
`whos`

Q-30:

You can get the value of a particular variable by typing its name  
`A`

Q-31:

`eigshow` Graphical demonstration of eigenvalues and singular values. `eigshow` presents a graphical experiment showing the effect on the unit circle of the mapping induced by various 2-by-2 matrices.

A pushbutton allows the choice of "eig" mode or "svd" mode.

In eig mode, the mouse can be used to move the vector  $x$  around the unit circle. The resulting trajectory of  $A*x$  is plotted. The object is to find vectors  $x$  so that  $A*x$  is parallel to  $x$ . Each such  $x$  is an eigenvector of  $A$ . The length of  $A*x$  is the corresponding eigenvalue.

In svd mode, the mouse moves two perpendicular unit vectors,  $x$  and  $y$ .

The resulting  $A*x$  and  $A*y$  are plotted. When  $A*x$  is perpendicular to  $A*y$ , then  $x$  and  $y$  are right singular vectors,  $A*x$  and  $A*y$  are multiples of left singular vectors, and the lengths of  $A*x$  and  $A*y$  are the corresponding singular values.

The figure title is a menu of selected matrices, including some with fewer than two real eigenvectors. eigshow(A) inserts A, which must be 2-by-2, in the menu.

```
eigshow(A)
```

Q-32:

```
% Define a matrix A.
A = [0 1 1 0 ; 1 0 0 1 ; 1 0 0 1 ; 0 1 1 0];
```

Q-33:

```
% Draw a picture showing the connected nodes.
cla
subplot(1,2,1);
gplot(A,[0 1;1 1;0 0;1 0],'.-');
text([-0.2, 1.2 -0.2, 1.2],[1.2, 1.2, -.2, -.2],('1234)', ...
    'HorizontalAlignment','center')
axis([-1 2 -1 2],'off')
```

Q-34:

```
% Draw a picture showing the adjacency matrix.
subplot(1,2,2);
xtemp=repmat(1:4,1,4);
ytemp=reshape(repmat(1:4,4,1),16,1);
text(xtemp-.5,ytemp-.5,char('0'+A(:)),'HorizontalAlignment','center');
line([.25 0 0 .25 NaN 3.75 4 4 3.75],[0 0 4 4 NaN 0 0 4 4])
axis off tight
```

Q-35:

**Here are the nodes in one hemisphere of the bucky ball, numbered polygon by polygon.**

```
subplot(1,1,1);
gplot(B(1:30,1:30),V(1:30,:),'b-')
for j = 1:30,
    text(V(j,1),V(j,2),int2str(j),'FontSize',10);
end
axis off equal
```

Q-36:

```
%The RAND command creates the matrix,
n = 10;
a = rand(n);
```

Q-37:

```
%the IMAGESC command plots an image of the matrix,
```



```
n = 10;  
a = rand(n);  
imagesc(a);  
colormap(hot); %automatically scaling the color map appropriately  
axis square;
```